

機器視覺檢測

光學元件規格及選用

教師：顏旭男





像機規格

- 相機規格
- 感光元件
- 取像快門
- 傳輸介面
- 相機接口
- 相機選用

相機規格

■一般相機：數位相機、單眼相機



相機規格

■一般相機：數位相機、單眼相機

➤工業相機

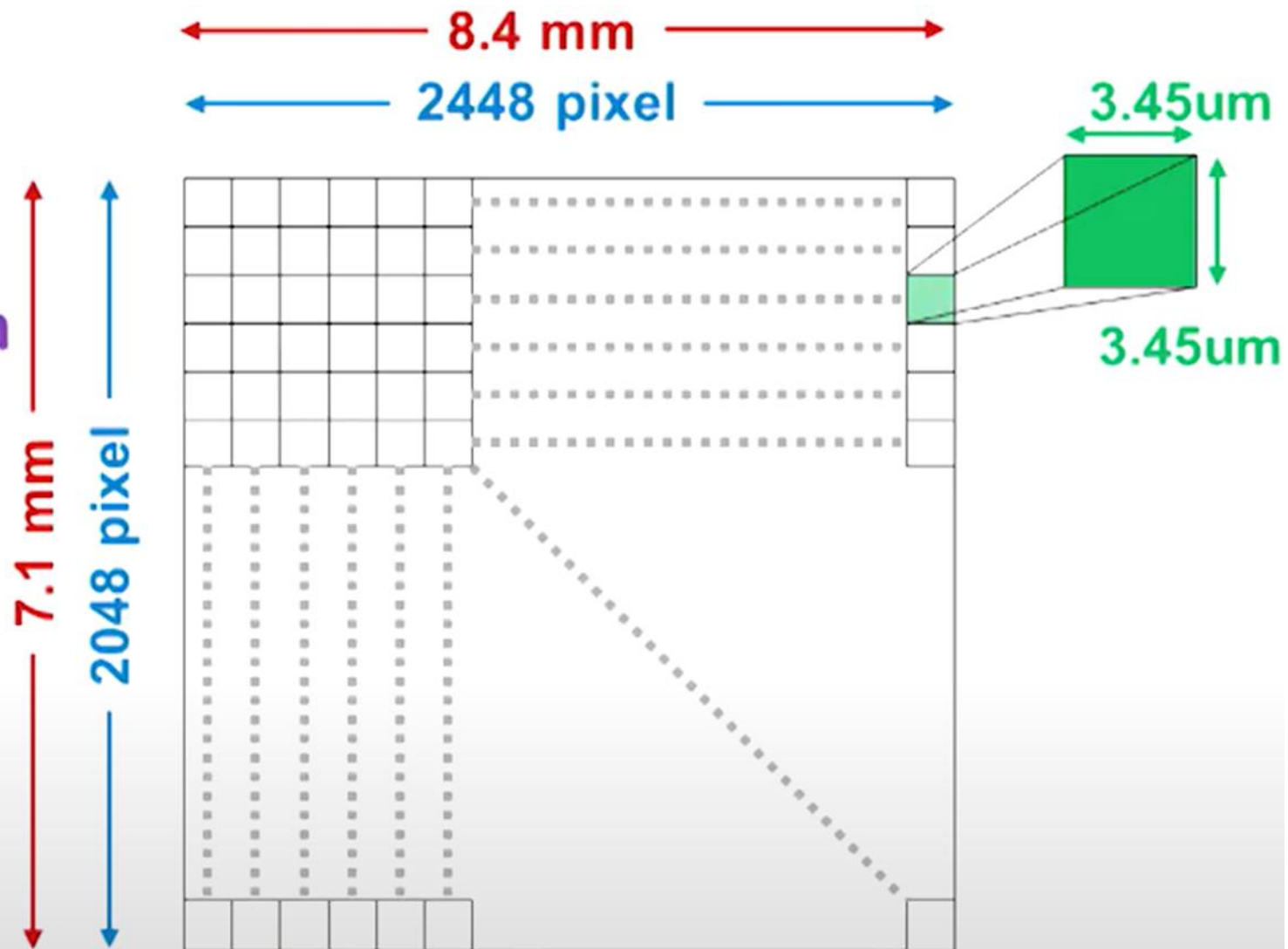
- 接口
- 感光元件
- 傳輸介面
- I/O與Power



感光元件

- 解析度=2448 x 2048
- Pixel Size=3.45um
- Sensor Size=8.4mm x 7.1mm

將此元件經鏡頭用來取像
(與 FOV 的關係)



感光元件

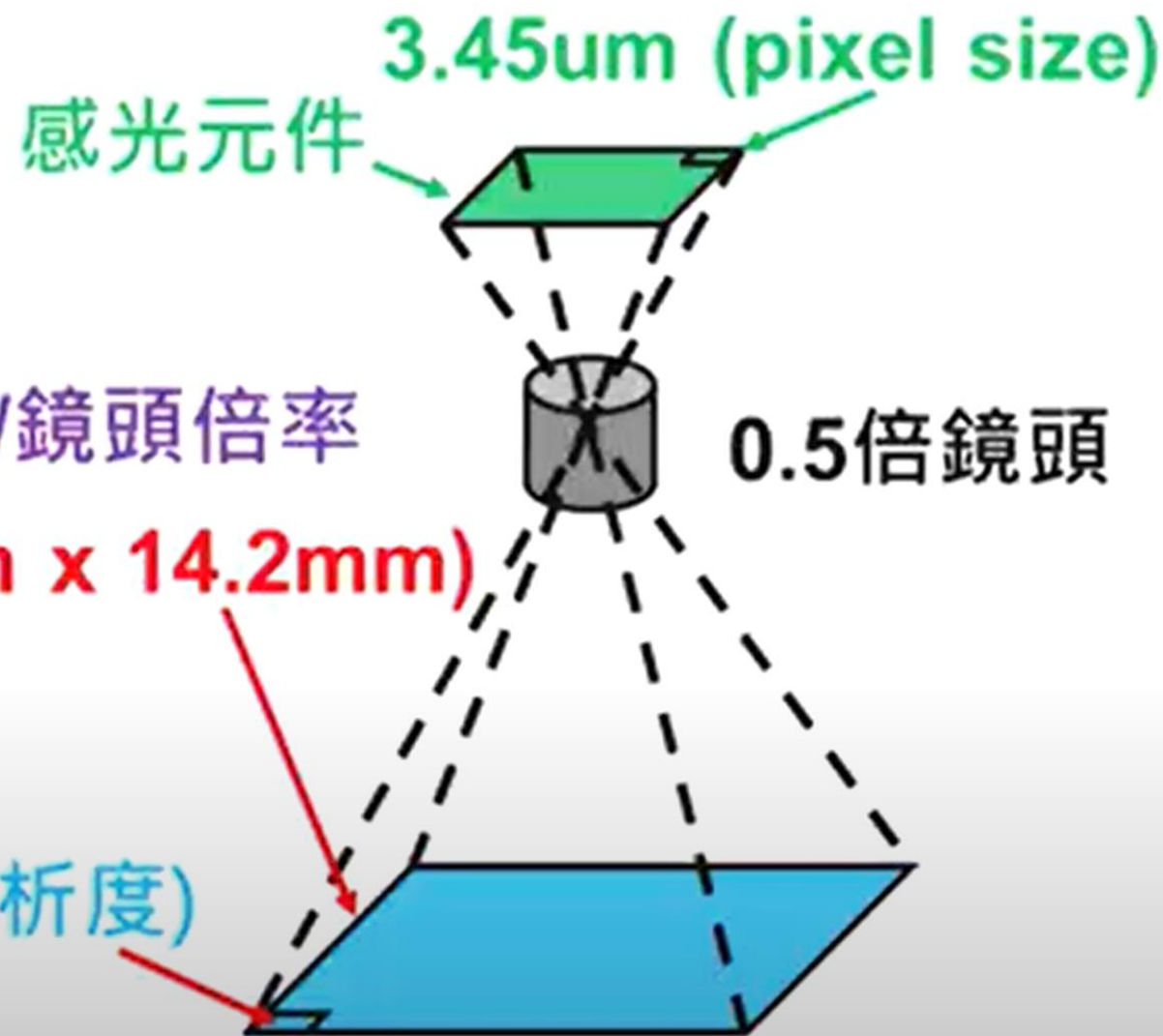
- 解析度=2448 x 2048
- Pixel Size=3.45um
- Sensor Size=8.4mm x 7.1mm
- 像素解析度(分辨率)=Pixel Size/鏡頭倍率

FOV(16.8mm x 14.2mm)

須提高分辨率?

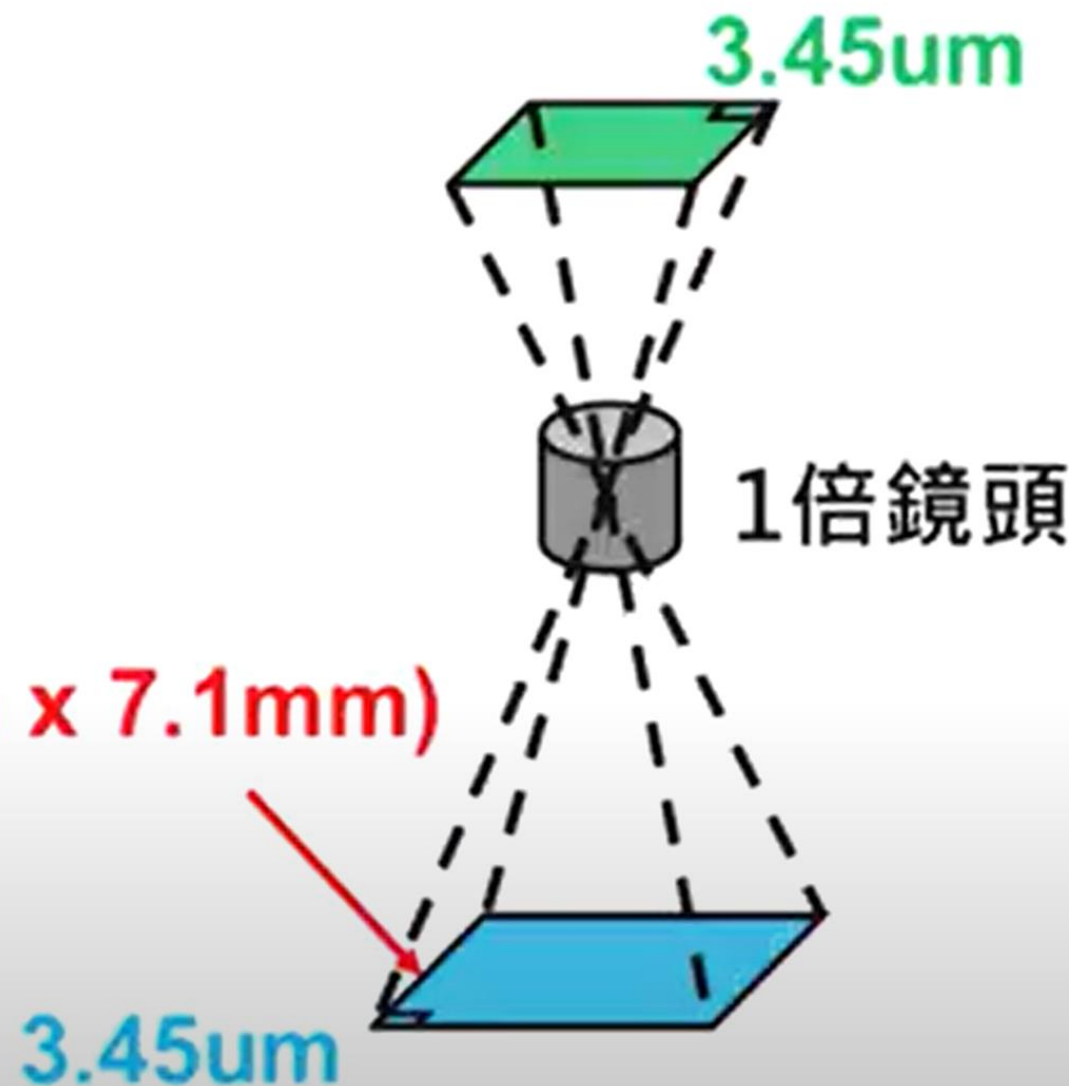


6.9um (像素解析度)



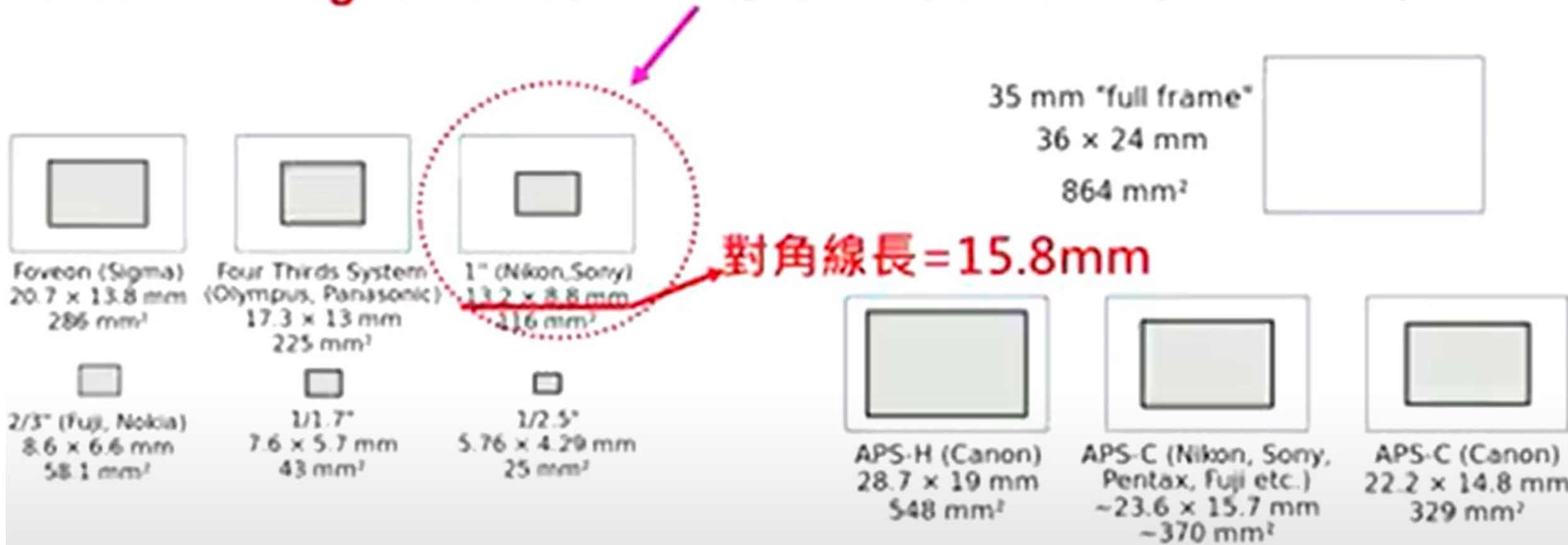
放大倍率

- 解析度=2448 x 2048
- Pixel Size=3.45um
- Sensor Size=8.4mm x 7.1mm
- 分辨率=Pixel Size/鏡頭倍率
 - 鏡頭倍率: 0.5 → 1
- FOV = 分辨率 x (解析度)
 - New FOV → 0.5 x FOV



感光元件

Q: 20mm image circle 的 lens 可用 1" 的 Sensor 嗎? Ans: 可以



取像快門

➤ Rolling Shutter與 Global Shutter比較



移動中的物體



旋轉中的物體

傳輸介面



GiG_E
VISION



US₃
VISION



CAMERA
Link



CoaPress
The next generation digital interface

(內含電源跟I/O)

傳輸介面

➤ 比較

名稱	傳輸距離	傳輸速度
GigE網路線	100m	1G bps
USB	10m	USB3.0 5Gbps USB2.0 480Mbps
Camera link	6m	2~6.8 Gbps
CoaXpress	Upto 68m	1.25~25 Gbps

相機接口

C-mount
CS-mount



F-mount



M42, M X X



相機選用

➤ <https://www.baslerweb.com/en/products/cameras/>

Get to know more about our camera portfolio on this page. We split the information into four parts: area scan cameras, line scan cameras, the world's leading developers and manufacturers of digital cameras for industrial image processing, we look back proudly on more than 2

Area Scan Cameras



Basler ace classic

Broad sensor selection, low priced, high performance



Basler ace U

Advanced feature set, latest CMOS sensor technology



Basler ace L

High resolution Sony Pregius 9 & 12 MP sensors up to 1.1", advanced feature set

相機選用

Camera model	Resolution (H x V)	Resolution	Sensor	Frame Rate	Mono/ Color	Interface
acA2500-14gm	2592 px x 1944 px	5 MP	MT9P031	14 fps	Mono	GigE
acA2500-14um	2592 px x 1944 px	5 MP	MT9P031	14 fps	Mono	USB 3.0
acA3800-10gc	3840 px x 2748 px	10 MP	MT9J003	10 fps	Color	GigE
acA3800-10gm	3840 px x 2748 px	10 MP	MT9J003	10 fps	Mono	GigE
acA3800-14uc	3840 px x 2748 px	10 MP	MT9J003	14 fps	Color	USB 3.0
acA3800-14um	3840 px x 2748 px	10 MP	MT9J003	14 fps	Mono	USB 3.0
acA4600-7gc	4608 px x 3288 px	14 MP	MT9F002	7 fps	Color	GigE
acA4600-10uc	4608 px x 3288 px	14 MP	MT9F002	10 fps	Color	USB 3.0

acA3800-14uc: 38xx H 解析度, 14fps (USB, Color), g: GigE, m: mono, k: camera link

USB 3.0 is faster than GigE

相機選用

acA3800-14uc

Camera Data

Interface	USB 3.0
Pixel Bit Depth	12 bits
Synchronization	▪ software trigger

Sensor Vendor	ON Semiconductor
Sensor	MT9J003
Shutter	Rolling Shutter
Max. Image Circle	1/2.3"
Sensor Type	CMOS
Sensor Size	6.4 mm x 4.6 mm
Resolution (H x V)	3840 px x 2748 px
Resolution	10 MP
Pixel Size (H x V)	1.67 μ m x 1.67 μ m
Frame Rate	14 fps
Mono/ Color	Color

相機選用

acA2000-50gm

Camera Data

Interface	GigE
Pixel Bit Depth	12 bits
Synchronization	▪ hardware trigger ▪ free-run ▪ Ethernet connection

Sensor Vendor	CMOSIS
Sensor	CMV2000
Shutter	Global Shutter
Max. Image Circle	2/3"
Sensor Type	CMOS
Sensor Size	11.3 mm x 6 mm
Resolution (H x V)	2048 px x 1088 px
Resolution	2 MP
Pixel Size (H x V)	5.5 μ m x 5.5 μ m
Frame Rate	50 fps
Mono/ Color	Mono

相機選用

acA3800-14uc	acA2000-50gm
3840 x 2748	2048 x 1088
10MP	2MP
Rolling Shutter	Global Shutter
14 fps	50 fps
Color	Mono
USB3.0	GigE



註記及感謝

**註記：本講義摘錄自YouTube中DELTA MOOCx
[科大]二維自動化光學檢測及應用影片，僅
作為教學輔助使用。**

**感謝台達電子工業股份有限公司製作分享，以及
雲科大吳先晃老師及北科大陳金聖老師精彩教學**